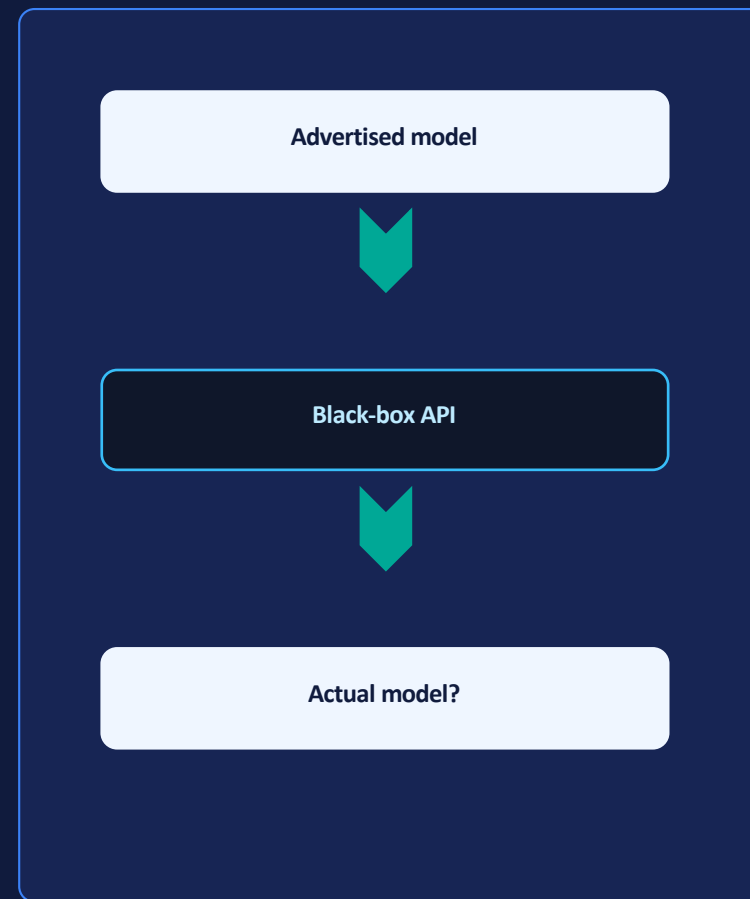


Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs:
Why Software Tests Fail and Hardware Attestation Wins

Commercial APIs create a trust gap: customers pay for named models, but black-box inference hides whether the advertised weights are actually served.



The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Threat model: black-box APIs create uncertainty over which distribution actually generated the response.

Formal audit question

Given samples from the API, can an auditor distinguish the specified distribution from the actual served distribution?



Economic driver

Cheaper serving paths reduce GPU costs while maintaining pricing for named models.



**Smaller
model**

8B instead
of 70B



**Quantized
variant**

FP8 / INT8
instead of FP16



**Fine-tuned
or updated**

Same family,
different weights



**Different
family**

Architecture
swap

Substitution types documented in the brief

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Benchmark comparison of original precision vs. FP8 quantized variants across five model families.

Model family	Precision	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	Original	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct	FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	Original	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct	FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	Original	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it	FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	Original	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct	FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	Original	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3	FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Signal problem

FP8 scores often remain within error bars, especially on MMLU and GSM8K.

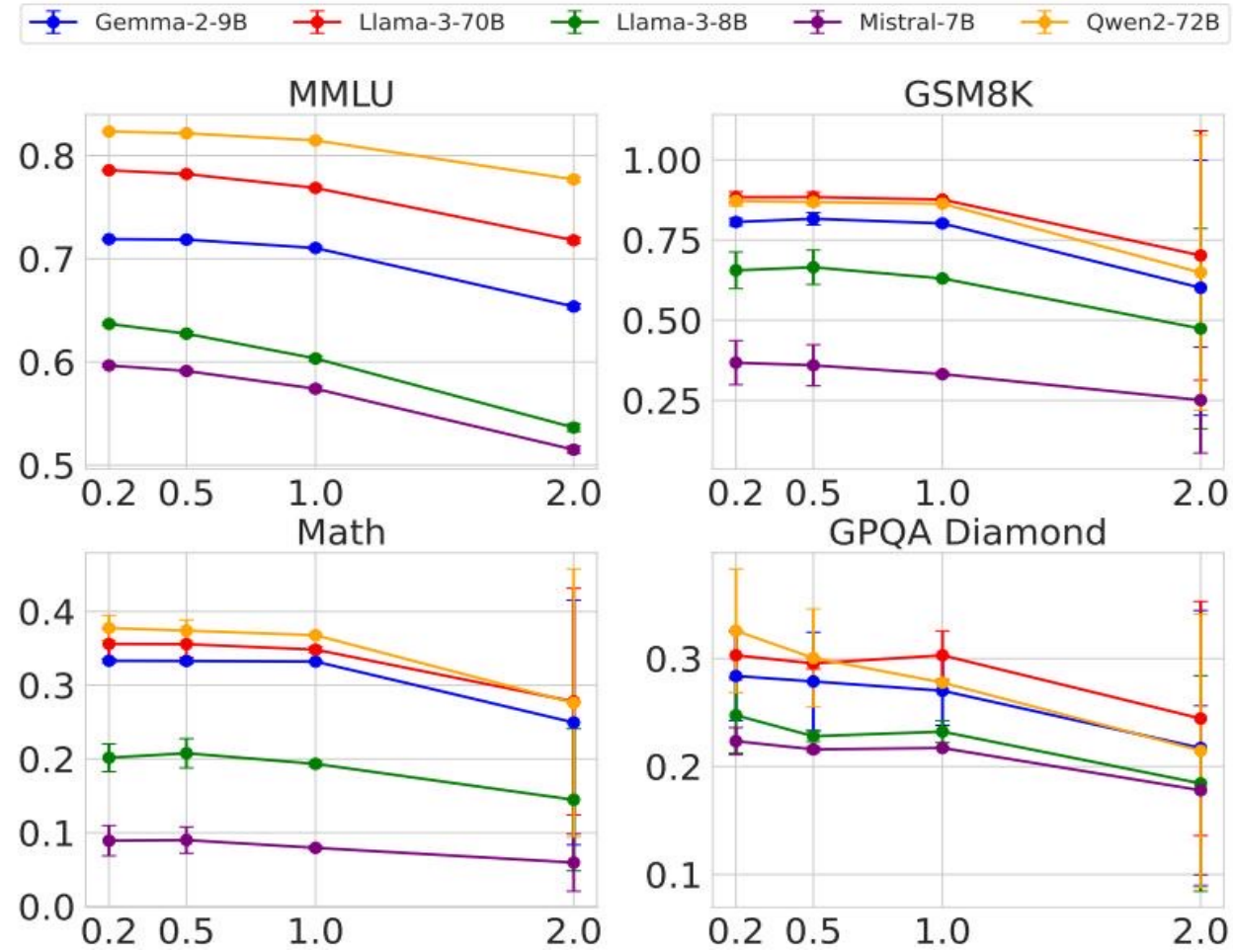


hard to audit

Values reproduced from source brief; ± denotes reported uncertainty.

Higher Temperature Amplifies Performance Gaps but Increases Noise

Temperature scaling experiments at 0.2, 0.5, 1.0, and 2.0 further obscure substitution signals.



Higher temperature degrades all models

At temperature 2.0, performance falls across tasks, adding noise to output-based comparisons.

Signal narrows or becomes noisy

The detection problem is not just low signal; sampling conditions can alter the measured gap.

Implication for auditors

Statistical comparisons must separate model identity from decoding configuration and stochasticity.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Same Llama-3-8B model, different production stack: log probabilities shift across frameworks and GPUs.

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure shows variation across transformers 4.40 / 4.50, vLLM 0.6.2 / 0.8.2, and A100 / H100 settings.

1 Reference
Local token-level log probabilities



2 API
Observed log probabilities



3 Mismatch
Infrastructure variation or substitution?

Auditor cannot reliably distinguish legitimate stack drift from model substitution.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

A provider can route only a fraction of traffic to a substitute, pushing the observed distribution toward the specified model.

Mixed serving distribution

$$P_{\text{mix}}(y|x) = (1 - p) \cdot P_{\text{spec}}(y|x) + p \cdot P_{\text{sub}}(y|x)$$



As p approaches 0, the mixed distribution approaches the specified model distribution, reducing the observable signal.

Adversarial audit surface



Randomized substitution

Serve the cheaper model only on a fraction p of queries.



Benchmark evasion

Detect known benchmark prompts and route them to the genuine model.



Practical effect

Software-only tests become query-prohibitive under active evasion.

Software-Only Auditing Methods Are Fundamentally Unreliable

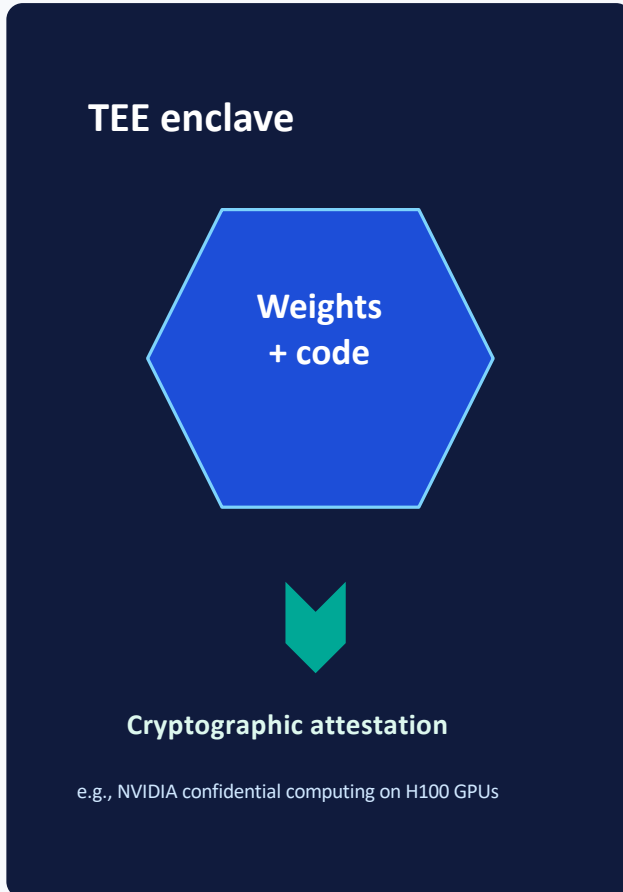
Across output statistics, log-probabilities, and active evasion, the detection surface remains brittle.

Method type	Weakness	Practical limitation
Text-based statistical tests (e.g., MMD)	Subtle substitutions and randomized routing reduce the observable output-distribution signal.	Impractically large query budgets; fail against subtle or randomized substitutions.
Log-probability fingerprinting	Production inference nondeterminism shifts token log probabilities across software versions and GPU hardware.	Cannot distinguish legitimate infrastructure differences from actual model substitution.
Adversarial countermeasures	Providers can use benchmark evasion or randomized substitution to target the audit protocol.	Known benchmark prompts can be routed to the genuine model; software-only detection becomes query-prohibitive.

Bottom line: software tests observe symptoms; they do not attest to loaded weights or executed inference code.

Trusted Execution Environments Provide Cryptographic Model Integrity

TEEs move verification below the software audit surface: prove the code and weights being executed.



Provable

Attests that specific model weights are loaded.



Efficient

Practical today with only modest performance overhead.



Robust

Verifies execution instead of inferring identity from outputs.

Why not only ZKPs?

The brief contrasts TEEs with zero-knowledge proofs: ZKPs are computationally prohibitive for large models, while TEEs are practical and deployable today.

Security shift

From statistical suspicion → cryptographic evidence.

Key Takeaways

Close the verification gap with hardware attestation

1 Economic motive

Black-box APIs enable covert swaps to cheaper serving paths.

2 Benchmarks blur

FP8 quantization barely affects many benchmark scores.

3 Metadata drifts

Log-prob fingerprints break across frameworks and GPU hardware.

4 Adversaries adapt

Randomized routing and benchmark evasion undermine software audits.

5 Attestation wins

TEEs prove loaded weights and executed code with deployable hardware support.

Call to action

Treat LLM API integrity as an infrastructure-verification problem, not a benchmark-detection problem.

Community priorities: attested inference APIs, verifiable deployment pipelines, and transparent integrity guarantees.